BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Records and Structured Data

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. How is a cell selected in a two-dimensional array?

2. What is a router's main job?
   A) Store user passwords
   B) Print network documents
   C) Run spreadsheet formulas
   D) Link different networks

3. "An IPv4 address is always six pairs of hexadecimal digits separated by colons." Correct this. Explain why.

4. Tablets need to join a wired LAN. "A WAP changes domain names into IP addresses." Correct this. How does a WAP help the tablets?

Today's objectives

Group related fields within a record. Use the technique in an independent scenario.

Key vocabulary - Read, cover, recall

Record

Groups related fields about one item or entity, potentially with different data types.

Field

A field is one named property within a record such as name or age.

Data types

A pupil record might store a string name, integer age and real score.

1D array versus record

An array stores indexed values of one type; records group named fields of potentially different types.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Record
Groups related fields about one item or entity, potentially with different data types. Apply this in a pupil profile.

Field
A field is one named property within a record such as name or age. Apply this in a library book.

Data types
A pupil record might store a string name, integer age and real score. Apply this in a sports member.

1D array versus record
An array stores indexed values of one type; records group named fields of potentially different types. Apply this in a patient appointment.

I Do - Worked model

A pupil profile is being developed. Explain how record and field are used. [4 marks]

- Read the scenario and identify the data the program needs.
- Record [1]; Groups related fields about one item or entity, potentially with different data types. [1].
- Field [1]; A field is one named property within a record such as name or age. [1].
- Two linked points for each concept.

We Do - Guided practice

1. A library book uses record. Explain how it helps. [2 marks]

2. Explain field in a sports member. [2 marks]

3. A patient appointment uses data types and 1d array versus record. Explain both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Each row is a pupil record: column 0 is the name; column 1 is the tutor group. Complete the index to print the second pupil's name.

```
array pupils[2,2]
pupils[0,0] = "Ada"
pupils[0,1] = "10A"
pupils[1,0] = "Ben"
pupils[1,1] = "10B"
print(pupils[...])
```

You Do - Independent exam questions

Q2. State TWO relevant techniques from this topic. [2 marks]

Q3. Explain how field and data types help with a train passenger. [4 marks]

Q4. A programmer designs a school register. Explain why record, data types and 1d array versus record matter. [6 marks]

Exit Ticket - True or False?

A record groups related fields about an item, even when the fields have different data types.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Use two indexes: one for the row and one for the column.

2. D - Link different networks

3. IPv4 supports addressing and routing. It is usually written as four decimal numbers separated by dots.

4. A wireless access point connects wireless devices to a network. It has a different job from a switch or router.

I Do - Explain the model

Read the scenario and identify the data the program needs.

Record [1]; Groups related fields about one item or entity, potentially with different data types. [1].

Field [1]; A field is one named property within a record such as name or age. [1].

Two linked points for each concept.

We Do - Model answers

1. Record [1]; Groups related fields about one item or entity, potentially with different data types. [1].

2. Field [1]; A field is one named property within a record such as name or age. [1].

3. Data types [1]; A pupil record might store a string name, integer age and real score. [1]. 1D array versus record [1]; An array stores indexed values of one type; records group named fields of potentially different types. [1].

Code First - Answer

```
array pupils[2,2]
pupils[0,0] = "Ada"
pupils[0,1] = "10A"
pupils[1,0] = "Ben"
pupils[1,1] = "10B"
print(pupils[1,0])
```

Expected output: Ben

You Do - Self-mark

Q2: Record [1]; Data types [1].

Q3: Field [1]; A field is one named property within a record such as name or age. [1]. Data types [1]; A pupil record might store a string name, integer age and real score. [1].

Q4: Record [1]; Groups related fields about one item or entity, potentially with different data types. [1]. Data types [1]; A pupil record might store a string name, integer age and real score. [1]. 1D array versus record [1]; An array stores indexed values of one type; records group named fields of potentially different types. [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.